
6 Mid-term Research Presentation

New Description

Outcome	Weight
Explore of IT Areas	◆◆◆◆

Outcome	Weight
Context of Research Project	◆◆◆◆◇

Outcome	Weight
Communication of Research Outcomes	◆◆◆◆

Outcome	Weight
Validation of Objectives and Methodology	◆◆◆◆

Outcome	Weight
Application of Skills and Knowledge	◆◆◆◆◇

Date	Author	Comment
2025/08/19 01:19	N/a Nikhil	Working On It
2025/08/19 14:23	N/a Nikhil	need 1 day to prepare the attack video.
2025/08/19 16:26	N/a Nikhil	Ready to Mark
2025/08/19 16:26	N/a Nikhil	Ready to Mark
2025/09/07 10:06	Ananda Maiti	Complete
2025/10/02 08:26	Ananda Maiti	Complete

DEAKIN UNIVERSITY

RESEARCH TRAINING AND PROJECT

ONTRACK SUBMISSION

Mid-term Research Presentation

Submitted By:

N/a NIKHIL

s224234862

2025/08/19 16:26

Tutor:

Ananda MAITI

Outcome	Weight
Explore of IT Areas	◆◆◆◆◆
Context of Research Project	◆◆◆◆◇
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Validation of Objectives and Methodology	◆◆◆◆◆
Application of Skills and Knowledge	◆◆◆◆◇

August 19, 2025





SIT 723

Research Techniques & Applications

Task 3.5D: Mid-term Research Presentation

Student Information

Name: Nikhil

Student ID: 224234862

Unit: SIT 723 - Research Techniques &
Applications

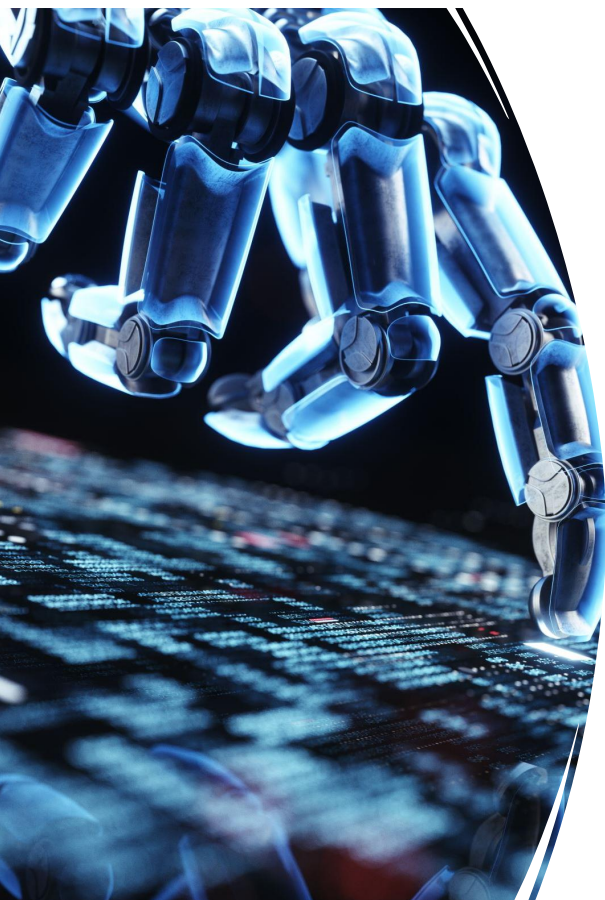


Video Presentation Link: <https://shorturl.at/kBDII>

Comment:

As part of Task 3.5D, I have attached my latest project presentation slides and also provided the video link. This work summarizes my progress, with particular focus on the implementation and challenges encountered. I look forward to constructive feedback.

Submission Date: August 19, 2025



Detection and Simulation of GPS Spoofing in Mobile Robots Using Sensor Cross-Validation

Nikhil (Student ID: 224234862)

SIT723 - Mid-term Research Presentation

Deakin University

August 2025



Research Aim

Primary Objective

Develop a real-time GPS spoofing detection system for autonomous mobile robots using sensor cross-validation in ROS-Gazebo-PX4 simulation environment

Key Goals

Dynamic GPS Trust Index for reliability quantification

Lightweight sensor fusion algorithms

Reproducible testing framework

>90% detection rate with <5% false alarms

Literature Review & Research Gaps

Key Literature Themes
GPS vulnerabilities in civilian signals
Sensor fusion detection methods (80-95% accuracy)
ROS-Gazebo-PX4 simulation frameworks
Trust-based mitigation strategies

Critical Research Gaps
Limited real-time validation on embedded platforms
Lack of standardized trust quantification
Gap between simulation and real-world deployment

Research Questions

RQ1: GPS Vulnerabilities

What are the primary vulnerabilities of GPS in autonomous mobile robots to spoofing attacks?

RQ2: Sensor Fusion Detection

How can GPS-IMU-odometry cross-validation effectively detect spoofing in real-time?

RQ3: GPS Trust Index

How can a dynamic GPS Trust Index quantify data reliability for autonomous navigation decisions?

Research Methodology

Integrated Sensor Cross-Validation Framework

Core Components

GPS-IMU-odometry cross-validation with adaptive thresholds

Dynamic GPS Trust Index based on sensor consistency

Statistical threshold-based anomaly detection

Implementation Platform

ROS-Gazebo-PX4 SITL simulation environment

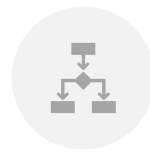
MAVROS bridge for real-time communication

Custom attack models: drift, jump, oscillation

Current Trimester Plan (2025)



WEEK 1-2: LITERATURE
REVIEW SYNTHESIS ?



WEEK 3-4: ROS-GAZEBO-PX4
ENVIRONMENT SETUP



WEEK 5-6: GPS-IMU-
ODOMETRY CROSS-
VALIDATION
IMPLEMENTATION



WEEK 7-8: DYNAMIC GPS
TRUST INDEX DEVELOPMENT



WEEK 9-10: SPOOFING
ATTACK SIMULATION NODES



WEEK 11-12: ALGORITHM
TESTING AND OPTIMIZATION

Future Work



2026



Comprehensive
performance
evaluation



Real-world
hardware
validation testing



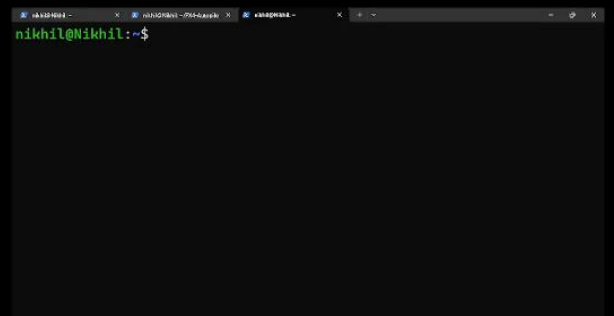
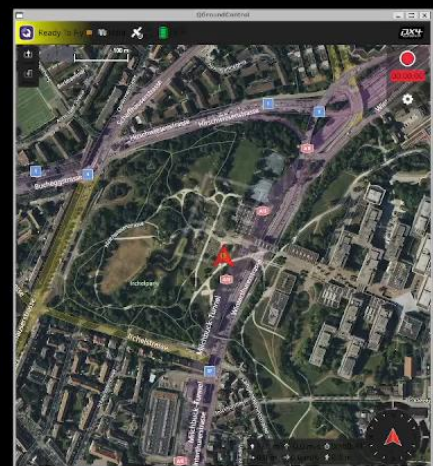
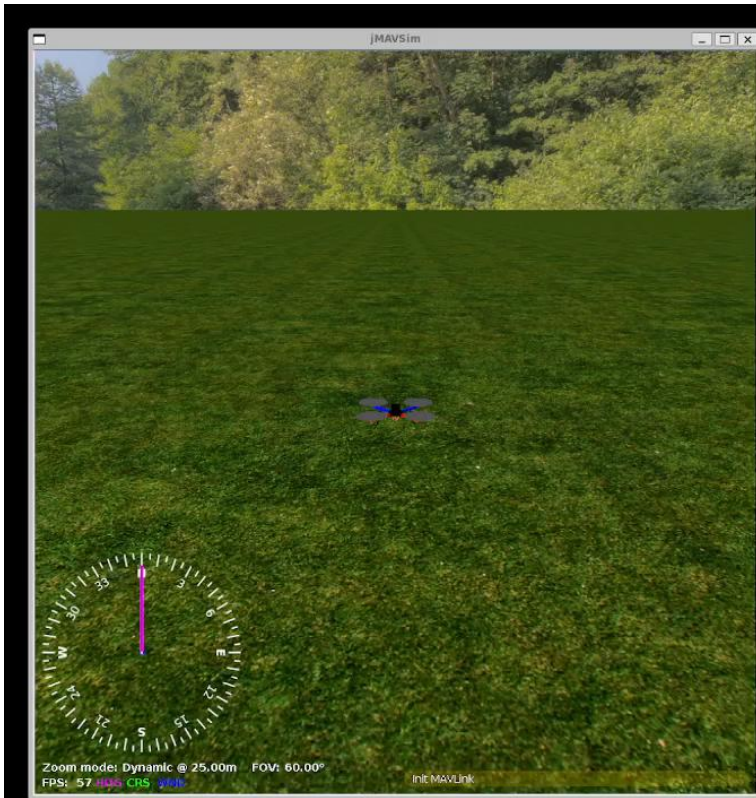
Trust Index
optimization and
tuning



Thesis
documentation and
writing



Results analysis and
publication
preparation



Expected Contributions



Novel Research Contributions



1. Dynamic GPS Trust Index for real-time reliability assessment



2. Lightweight detection framework for embedded robots



3. Reproducible simulation environment for GPS spoofing research



4. Optimized sensor cross-validation algorithms



5. Comprehensive evaluation metrics and benchmarking

An aerial photograph of a long, multi-lane highway bridge stretching across a body of water. The water is a vibrant turquoise color with visible ripples. The bridge has several lanes in both directions, with white lane markings. Several vehicles, including cars and trucks, are visible on the bridge. The sky is not visible, as the image is cropped to focus on the bridge and water.

Thank You

Presentation By Nikhil

Detection and Simulation of GPS Spoofing in Mobile Robots
Using Sensor Cross-Validation